



[Products](#) / [DataX](#) / [DataX Manual](#) / [Introduction](#)

Introduction

Introduction

- What is DataX?
- Key Capabilities
- Why DataX Matters
- Prerequisites
- Getting Started
- Typical Scenario
- Architecture Overview
 - High-Level View
 - Core Components
 - Ingestion Lifecycle
- Who Should Use This Manual

What is DataX?

DataX is the SAYA Platform's ingestion layer that brings data in from external sources (files, APIs, databases, email, object storage) and delivers it into internal data sets for downstream processing.

It is designed to be a **bidirectional data gateway**:

- **Inbound**: External source → DataX → Data Sets
- **Outbound** (where applicable): Data Sets → DataX → External targets

Key Capabilities

Capability	Description	Business Value
Multi-source connectivity 🔄	Supports SFTP, HTTP, Database, Email, Object Storage connectors with reusable authentication	Connect once, use across multiple data formats and ingestion jobs
Format flexibility	Handles CSV, Excel, JSON, XML, SWIFT MT/MX formats with configurable field mappings	Process diverse file types without custom code

Capability	Description	Business Value
Automated scheduling	Cron-based polling for regular data extraction from external systems	Eliminate manual file downloads and reduce operational overhead
Validation & quality checks	Pre-ingestion validation for file name patterns, schema compliance, and required fields	Catch data issues before processing, reducing downstream errors
Execution tracking	Complete audit trail with status, timestamps, record counts, and error details	Full transparency for compliance and troubleshooting

Why DataX Matters

Reason	Business Impact	ROI Indicator
Standardised ingestion	Avoids bespoke one-off data import logic per client/source	↓ Integration time
Data quality & consistency	Enforces formats, validations, and schema alignment	↓ Reprocessing
Operational visibility	Ingestion history, statuses, and execution details improve auditability	↑ Transparency
Reusable configurations	Connectors/formats are reusable across multiple datasets	↓ Setup effort

Prerequisites

Before using DataX, ensure:

- **System Access:** Appropriate role permissions for Operations User or Client Admin
- **External System Details:** Connection credentials, host/port information, or SFTP/API endpoints for data sources
- **Data Understanding:** Knowledge of incoming file formats, field structures, and expected data volumes
- **Configuration Authority:** For admins setting up connectors, formats, and datasets

Getting Started

Role	Start Here	What You'll Do
Operations User	Workspace Guide	Upload files, monitor ingestion status, investigate execution details

Role	Start Here	What You'll Do
Client Admin	Configurations	Set up Data Connectors, create Data Formats, configure Data Sets
New to DataX?	Use Cases	Learn through real-world scenarios and end-to-end workflows

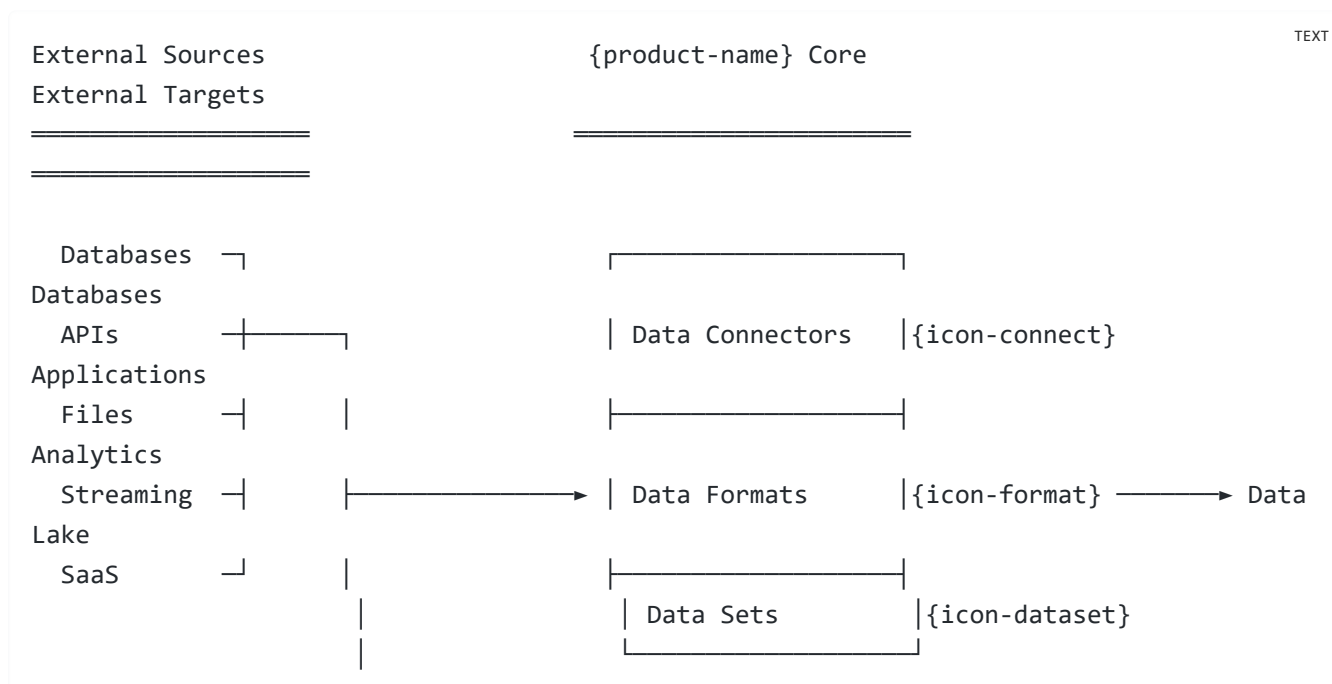
Typical Scenario

A client uploads a daily bank statement file:

1. Admin configures a **Data Connector** for the source
2. Admin creates a **Data Format** that maps incoming fields
3. Admin creates a **Data Set** and links the format
4. Operations user uploads the file via **Manual Ingestion** 
5. DataX validates and processes the file and records an execution in **Ingestion History**

Architecture Overview

High-Level View



Core Components

Component	Purpose	Examples
Data Connectors 	Connectivity + authentication to external systems. Reusable across multiple formats and sets.	OAuth2, API Key, DB connection pooling, credential storage
Data Formats	Defines how incoming data is read and transformed. Includes mapping and validation.	CSV/JSON/XML parsing, field mapping, type conversion
Data Sets 	Logical organisation of ingested data for business use.	Transaction dataset, customer dataset

Ingestion Lifecycle

1. **Identify:** File/source arrives with a **Source Identifier**
2. **Validate:** Format detection + schema validation
3. **Execute:** Ingestion execution is created and processed
4. **Observe:** Status & metrics appear in Ingestion History
5. **Resolve:** Failures/partials can be investigated using Execution Details

Who Should Use This Manual

Role	Focus Sections	Quick Start
Operations User	Workspace, Manual Ingestion, Ingestion History	Workspace
Client Admin	Configurations (Connectors, Formats, Data Sets)	Configurations
Support / Ops	Troubleshooting, Execution Details	Troubleshooting



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